

Yidong MING

Embodied AI | Robot Learning | Vision-Language-Action Models | LLM Agents

Hong Kong | yidonming2-c@my.cityu.edu.hk | +852 5197 2928

Google Scholar

Education

City University of Hong Kong, BSc in Data Science (expected Jun 2027) Sep 2023 – Jun 2027
CGPA: 4.08/4.3 | Rank: Top 2% of cohort | Dean's List: All semesters

Awards & Honors

- **HKSAR Government Scholarship for Outstanding Performance** (HK\$80,000) 2025/26
Recommended by CityUHK for renewal in 2026/27.
- **HKSAR Government Scholarship Fund – Reaching Out Award** (HK\$10,000) 2025/26
- **Hong Kong, China – Asia-Pacific Economic Cooperation Scholarship** (HK\$80,000) 2025/26
- **Chi-Li Pao Foundation Scholarship for Academic Excellence** (HK\$25,000; one recipient in the College) 2023/24
- **Greater Bay Area Data Governance Competition – Grand Prize** (sole winner among nearly 100 teams) 2025
- **14th Asia-Pacific Mathematical Contest in Modeling (APMCM) – Second Prize** 2024
- **CityU Tiger Program – Top 10% in CityU**
- **Hua Tai Securities Investment Banking Internship Preparatory Camp – Second Place** 2024

Research Experience

Embodied AI, InfoBodied AI Lab, The University of Hong Kong Jun 2026 – Present
HKU Summer Research Program | Supervised by Prof. Yanchao Yang

- **Co-first author** of *Transferable Behavior Embeddings from Unlabeled Videos for Policy Learning*, which transfers behavior representations learned from unlabeled manipulation videos into robot policies through auxiliary supervision.
- Conducted the **representation-probing experiments**, covering ongoing skills, progress, completion transfer, and spatial readouts; evaluated what frozen behavior embeddings encode and how this information transfers across tasks.
- Ran the $\pi_{0.5}$ **simulation experiments in Table 1 and all experiments in Table 2**. Behavior supervision improved clean-condition mean success from **37.2% to 50.2%** across 11 RoboTwin 2.0 tasks, without additional inference cost.
- Contributed to **Mobile ALOHA experimental design and execution**. Across two unseen object–destination pairings, the behavior-supervised policy achieved **49/80** successful trials, compared with **0/80** for the baseline.
- **Third author** of *RoboCompiler: Agentic Experience Compilation for Robot Post-Training*; primarily responsible for **real-robot experimental design and execution**. Mean success across three physical ARX tasks improved from **75.0% to 88.3%**.

Efficient Inference for Diffusion Language Models, Final Year Project, City University of Hong Kong Sep 2026 – Present
Supervised by Prof. Jianyuan Guo

- Investigate **token-level sparsity and dynamic-depth inference** for LLaDA-8B-Instruct and DyLLM, focusing on reducing repeated computation while preserving generation quality.
- Built a **profiling and experiment-analysis framework** for active-token dynamics, cross-layer support persistence, spatial locality, block survival, and runtime/scheduler tracing.
- Designed and evaluated **Depth Stream** strategies. In preliminary MMLU-Pro measurements, Step246 Loose achieved a **1.46× generation-time speedup** relative to the DyLLM baseline, with scores of **37.67% versus 36.61%**.

Federated Learning & Spiking Neural Networks, City University of Hong Kong Apr 2026 – Jul 2026
Supervised by Maolin Wang & Xiangyu Zhao

- Initiated the federated SNN class-unlearning direction and co-designed **SpikeBypass**, a bypass-decoupled framework with a frozen backbone and a plug-and-play adapter patch.
- Co-designed **basis-aligned spike-saliency localization** and mask-constrained adapters that map temporally aggregated logical-channel activity to reused physical filter-bank coordinates; ran most ablation studies on suppression strength, saliency coverage, and retained-class utility.
- Across Fashion-MNIST, CIFAR-10, and CIFAR-100, the framework suppressed the forget class to **0% accuracy** while retaining **87.82%, 88.06%, and 33.68%** accuracy on the remaining classes, respectively.

- Investigated the integration of machine learning, large language models, and advanced control systems for industrial-process optimization and new-energy transformation.
- Automated industrial data-analysis workflows with function tools, workflow orchestration, agents, and the Model Context Protocol (MCP), while exploring PID/MPC strategies for operational and environmental objectives.

LLM Alignment & Evaluation, City University of Hong Kong

Nov 2025 – Jan 2026

Research collaboration with Junyi Zhou

- Co-authored **RubricBench**, a benchmark of **1,147 pairwise comparisons** for evaluating rubric-guided LLM judges, curated across chat, instruction-following, STEM, coding, and safety domains.
- Contributed to multidimensional hard-sample filtering and instruction-derived atomic rubric annotation; experiments exposed a **27-percentage-point gap** between human- and model-generated rubrics.

Publications & Manuscripts

- **Transferable Behavior Embeddings from Unlabeled Videos for Policy Learning**
Ruizhe Liu*, **Yidong Ming***, Zewen Wu, Qian Luo, Jianyuan Guo, Yanchao Yang. (* equal contribution)
ICLR 2027 – under review
- **RoboCompiler: Agentic Experience Compilation for Robot Post-Training**
Qian Luo, Wentao Guo, **Yidong Ming**, Zhangke Liang, Haijian Tang, Yi Ma, Yanchao Yang.
ICLR 2027 – under review
- **SpikeBypass: Bypass-Decoupled Federated Class Unlearning for Spiking Neural Networks**
Haodong Huo*, **Yidong Ming***, Maolin Wang. (* equal contribution)
arXiv preprint
- **RubricBench: Aligning Model-Generated Rubrics with Human Standards**
Junyi Zhou, Qiyuan Zhang, Yufei Wang, Fuyuan Lyu, **Yidong Ming**, et al.
ACL 2026 Main Conference
- **CheMatAgent: Enhancing LLMs for Chemistry and Materials Science through Tree-Search Based Tool Learning**
Mengsong Wu, YaFei Wang, **Yidong Ming**, et al.
arXiv:2506.07551

Technical Skills

Robot Learning & Embodied AI: Vision-Language-Action Models, Policy Learning, Imitation Learning, Multimodal Representation Learning, Cross-Embodiment Generalization**Robotics & Vision:** ROS 2, MuJoCo, RoboTwin 2.0, OpenCV**Programming & Machine Learning:** Python, PyTorch, JAX, NumPy, CUDA, Git, Linux**Additional Research:** LLM Agents, Tool Calling, Self-Supervised Learning, Federated Learning, Spiking Neural Networks**Engineering Experience**

Embodied AI Algorithm Intern, TransEngram

Aug 2026 – Present

Shenzhen, China

- Conduct research on **label-free manipulation concept learning and vision-language-action models**, extending representations learned from large-scale robot trajectories toward stronger policy learning and cross-embodiment generalization.
- Develop an **embodied-data closed-loop pipeline** for iterative robot data collection, curation, training, and evaluation, enabling continual improvement of embodied policies through real-world interaction data.

AI Research Engineer Intern, Green Dynamics Limited

Oct 2024 – Oct 2025

- Contributed to **CheMatAgent**, an LLM-based scientific agent integrating 137 external tools for chemistry and materials science, enabling autonomous tool selection, execution, and multi-step scientific workflows.
- Implemented **pymatgen tools for LLM tool calling**, standardizing materials-science functions, parameter schemas, documentation, and executable interfaces for end-to-end materials R&D workflows.
- Developed and evaluated machine-learning models and LLM-based applications to accelerate AI-driven materials R&D workflows, adapting NLP algorithms and profiling core modules.
- Translated client requirements into technical AI solutions, iterated workflows based on real-world feedback, and contributed to standardized AI toolkits and technical documentation.

Computer Vision Project Assistant, City University of Hong KongJun 2024 – Sep 2024

Industry collaboration with Chow Tai Fook Jewellery | Supervised by Prof. Raymond Chan & Prof. S. Yung

- Contributed to an **industrial computer-vision system for diamond-inclusion detection**, preparing image datasets, implementing preprocessing and augmentation pipelines, and training a **U-Net-based semantic-segmentation model** for pixel-level inclusion localization.
- Built a **Python image-retrieval tool** to match original diamond images, annotated images, and cropped inclusion regions for efficient dataset inspection and analysis.

Leadership & Community Service

- Served on the **Board of Undergraduate Studies** at City University of Hong Kong, contributing to undergraduate academic and student-affairs matters.
- Participated in the **Student Chapter of the Department of Data Science** at City University of Hong Kong, supporting departmental student activities.
- Volunteered with **Hong Kong Ferryman**, a volunteer organization serving CityU undergraduates.
- Served as a **volunteer teacher** and supported community fundraising through Flag Days organized by **Chinese Evangelical Zion Church** and **HOPE worldwide**.
- Assisted with the **SCHKM 2024 & Mainland China Alma Mater Program**.

Additional Experience

Student Teaching Assistant, City University of Hong KongSummer 2024

- Supported summer instruction for **MA1508 – Calculus** and **MA1503 – Linear Algebra with Applications** at City University of Hong Kong.

Office Assistant, Department of Data Science, City University of Hong KongFeb 2024 – Aug 2025

- Provided administrative and data-processing support, including document management, desktop research, event coordination, visitor assistance, and telephone enquiries.

Student Ambassador, Shanghai Golden EducationDec 2023 – Feb 2024

- Managed WeChat and Xiaohongshu content, supported online and offline events, conducted student interviews, evaluated products, and analyzed customer data for targeted outreach.

Digital Marketing Assistant, Guangzhou Saifu Information Technology Co., Ltd.Summer 2024

- Supported digital-marketing strategy, WeChat and social-media operations, influencer coordination, campaign reporting, and website-development planning for the “Counting Treasure” application.

Other Activities & Achievements

- 2nd City I&T Grand Challenge Semi-Pitch2024
- Hua Tai Securities Mock Investment Banking Department Roadshow – Second Prize
- Mathematical Contest in Modeling – S Prize
- Dormitory Jump Rope Competition – Second Place
- ICFD Badminton Competition – First Runner-up

Exchange Experience

Middlesex University, London, United KingdomMay 2026 – Jun 2026

Academic Exchange & IELTS Preparation Program

The University of British Columbia, Vancouver, CanadaJul 2025 – Aug 2025

Visiting Summer Program